

Restaurant Vibe Classification with Locally Fine-Tuned DistilBERT

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Abstract

1 Introduction

2 Problem Definition

2.1 Vibe Labels

Label	Description
study_spot	Quiet cafes and work-friendly spaces
late_night	Bars, nightlife, and evening atmosphere
casual_hangout	Relaxed and informal dining
family_friendly	Restaurants suited for families and groups
upscale	Elegant or fine dining experiences
trendy	Modern, stylish, or social restaurants
date_night	Romantic or intimate dining atmosphere

Table 1: Restaurant vibe classification labels

3 Dataset

3.1 Data Source

3.2 Dataset Filtering

3.3 Dataset Construction

3.4 Labeling Process

3.5 Train / Validation / Test Split

4 Model and Fine-Tuning Method

4.1 Model Selection

4.2 Tokenization

4.3 Fine-Tuning Method

4.4 Baseline Model

5 Experimental Setup

5.1 Training Configurations

Experiment	Learning Rate	Epochs	Notes
Model 1	3e-5	8	Baseline fine-tuning
Model 2	2e-5	8	Warmup scheduling
Model 3	3e-5	10	Extended training duration

Table 2: DistilBERT experiment configurations

6 Results

6.1 Final Model Comparison

Model	Accuracy	Weighted F1
TF-IDF + Logistic Regression	0.67	0.66
DistilBERT (Model 1)	0.67	0.68

Table 3: Baseline vs. fine-tuned DistilBERT performance

Input Summary	True Label	Predicted Label
Quiet cafe with students studying	study_spot	study_spot
Stylish cocktail lounge	trendy	date_night
Upscale wine bar	upscale	trendy
Family breakfast diner	family_friendly	family_friendly

Table 4: Example predictions from the final model

6.2 Confusion Matrix

7 Error Analysis

7.1 Qualitative Examples

8 Ethical and Practical Considerations

9 Conclusion